A Unified Medium-Based Cosmology：

—Resolving the Hubble Tension and Galactic Gravity with TEEM—

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Abstract

We present the TEEM framework, a medium-based cosmological model that unifies cosmic expansion and galactic dynamics through the evolution of an underlying ETR medium. In TEEM, the flow of time, the strength of gravity, and the expansion rate are governed by the density, depletion, and regeneration of this medium. We test TEEM against multiple independent observational probes, including the Pantheon+SH0ES Type Ia supernova sample (1590 SNe), BAO distance measurements, the CMB acoustic scale, galaxy rotation curves, and a simulated galaxy-collision scenario.

Using SN Ia data, TEEM fits the luminosity-distance relation with ΛCDM-level accuracy, yielding $H_0 = 69.58$, $\Omega_M = 0.326$, $\Omega_{\mathrm{ETR0}} = 0.771$, and a medium-regeneration index $\beta = 0.363$. The model naturally produces a high local Hubble constant ($H_0 = 78.88$) and a lower global value ($H_0 = 68.06$), offering a physical explanation for the Hubble tension through non-uniform medium density and time flow. The same SN-derived parameters reproduce BAO transverse distances and yield a CMB acoustic angle ($\theta_* = 0.011094$ rad) close to the ΛCDM prediction.

On galactic scales, TEEM explains flat rotation curves via medium depletion, quantified by a measurable parameter a_E. Fits to rotation-curve data give $a_E \approx 148.6$, reproducing dark-matter-like gravitational behavior without invoking dark matter halos. In a galaxy-collision simulation, $a_E$increases from 157.4 to 275.5, demonstrating that medium disruption enhances gravitational strength, consistent with observations of interacting galaxies.

Together, these results show that TEEM provides physical origins for phenomena traditionally attributed to dark matter and dark energy, while maintaining consistency with key cosmological observations. TEEM therefore emerges as a viable unified alternative to ΛCDM, linking cosmic acceleration, galactic gravity, and the Hubble tension to the dynamics of a single underlying medium.

1. Introduction

1.1 Overview of the ΛCDM model and its observational successes

- Introduce ΛCDM as the standard cosmological model.
- Mention its strong agreement with:
    - CMB anisotropies
    - BAO distance scales
    - Large-scale structure formation

  - SN Ia luminosity distances
- Emphasize that ΛCDM provides a simple and powerful description using only six parameters.
- State that despite its success, ΛCDM relies on two unknown components: **dark matter** and **dark energy**, which together constitute ~95% of the Universe.

Writing cue: This section sets up ΛCDM as successful but incomplete.

1.2 Persistent tensions

Introduce the three major observational tensions that motivate new physics.

1.2.1 The Hubble tension (local vs global H_0)

- Local measurements (Cepheids + SN Ia) give **$H_0 \approx$ 73–75 km/s/Mpc**.
- Early-Universe measurements (Planck CMB) give **$H_0 \approx$ 67.4 km/s/Mpc**.
- The discrepancy exceeds 5σ and cannot be resolved by simple parameter shifts.
- ΛCDM assumes a uniform cosmic time flow, which may be too restrictive.

1.2.2 Galaxy rotation curves and the dark matter problem

- Observed rotation curves remain flat at large radii.
- Baryonic mass alone cannot explain the velocities.
- ΛCDM introduces dark matter halos, but their profiles often disagree with observations:
  - Core–cusp problem
  - Diversity problem
  - Too-big-to-fail problem
- MOND reproduces some trends but lacks cosmological grounding.

1.2.3 Enhanced gravitational signatures in interacting galaxies

- Interacting and merging galaxies often show **stronger-than-expected gravitational potentials**.
- Examples: M51, Antennae, Mice galaxies.
- ΛCDM requires rapid dark matter redistribution, which is physically unclear.
- Observations suggest **environment-dependent gravity**, hinting at medium-based effects.

Writing cue: This section establishes that multiple independent anomalies point toward missing physics.

1.3 Limitations of existing alternatives

Discuss why current alternatives cannot fully resolve the tensions.

- **MOND:**
  - Explains rotation curves but fails for clusters and cosmology.
  - No physical mechanism for acceleration or CMB peaks.
- **Modified gravity (f(R), scalar–tensor, etc.):**
  - Often conflict with solar-system tests.
  - Hard to reconcile with CMB + BAO simultaneously.

- **Exotic dark energy models:**
    - Add parameters but do not explain galaxy-scale anomalies.
    - Typically phenomenological rather than physical.

Writing cue: This section motivates the need for a model that works on *both* cosmic and galactic scales.

1.4 Motivation for a medium-based cosmological framework

Introduce the conceptual foundation of TEEM.

- Propose that spacetime is permeated by an **ETR medium** whose density and dynamics govern:
    - The flow of time
    - The strength of gravity
    - The rate of cosmic expansion
- Medium depletion enhances gravity (galactic dynamics).
- Medium regeneration drives cosmic acceleration (dark energy analogue).
- Non-uniform medium density produces different effective H_0 values (Hubble tension).
- Interactions disturb the medium, explaining enhanced gravity in merging galaxies.

Writing cue: This section explains *why* a medium model is physically motivated.

1.5 Purpose of this work

Clearly state the goals of the paper.

- **Introduce the TEEM model** as a unified medium-based cosmological framework.
- **Test TEEM against multiple independent datasets:**
    - SN Ia (Pantheon+SH0ES)
    - BAO distance measurements
    - CMB acoustic scale
    - Galaxy rotation curves
    - Galaxy collision scenarios
- **Evaluate whether TEEM can simultaneously reproduce:**
    - Cosmic expansion history
    - Local vs global H_0
    - Dark-matter-like galactic dynamics
    - Collision-enhanced gravitational signatures
- **Assess whether TEEM can serve as a unified alternative to ΛCDM.**

Writing cue: This section ends the Introduction by stating the contribution of the paper.

2. The TEEM Framework

2.1 Concept of the ETR Medium

We introduce the *ETR medium* as a physical substrate that permeates spacetime and governs the flow of time, the strength of gravity, and the rate of cosmic expansion. In TEEM, spacetime is not treated as a static geometric manifold but as a dynamical medium whose density and evolution determine observable cosmological and galactic phenomena.

The ETR medium possesses three fundamental properties:

- **Medium density** ρ_{ETR}: Controls the local rate at which physical processes update. Regions with higher medium density experience a faster effective flow of time.
- **Medium depletion**: Matter locally consumes or disturbs the medium, reducing ρ_{ETR}. This depletion enhances gravitational effects, producing rotation-curve anomalies without invoking particulate dark matter.
- **Medium regeneration**: The medium is continuously replenished by a background process. The regeneration rate determines the cosmic acceleration and replaces the role of dark energy.

Physically, the ETR medium acts as an **information-update substrate**: the Universe evolves because the medium updates, and the rate of this update defines the passage of time.

2.2 Cosmological Dynamics

At cosmological scales, the dynamics of the ETR medium modify the expansion history of the Universe. The TEEM expansion law is given by:

$$H^2(z) = H_0^2\left[\Omega_M(1+z)^3 + \Omega_{\mathrm{ETR0}}(1+z)^{\beta}\right].$$

The first term corresponds to standard matter. The second term represents the contribution of the ETR medium, whose density evolves as a power law with index β.

Role of β

The parameter β quantifies the **medium regeneration rate**:

- $\beta = 0$: constant medium density
- $\beta > 0$: medium density increases toward higher redshift
- $\beta < 0$: medium density decreases toward higher redshift

Cosmic acceleration arises naturally when the regeneration term dominates at late times.

Relation to ΛCDM

ΛCDM is recovered as a special case:

$$\beta = 0, \Omega_{\mathrm{ETR0}} = \Omega_{\Lambda}.$$

Thus, TEEM generalizes ΛCDM by providing a **physical origin** for the dark-energy term.

2.3 Galactic Dynamics

At galactic scales, medium depletion modifies the effective gravitational potential. The TEEM rotational velocity law is:

$$v^2(r) = \frac{GM}{r} + a_E r.$$

The first term is the Newtonian contribution from baryonic matter. The second term arises from **ETR medium depletion** around matter concentrations.

Interpretation of $\boldsymbol{a_E}$

The parameter a_E is the **medium depletion coefficient**:

- Higher a_E corresponds to stronger medium depletion
- Depletion enhances gravity without requiring dark matter
- Interactions or collisions can increase a_E, producing stronger gravitational signatures

Physical meaning

Medium thinning reduces the local information-update capacity, effectively deepening the gravitational potential. This mechanism reproduces:

- Flat rotation curves
- MOND-like behavior
- Enhanced gravity in interacting galaxies
- Dark-matter-like effects without invoking new particles

2.4 Expected Observational Signatures

The TEEM framework predicts several observational phenomena that directly correspond to known astrophysical anomalies:

Non-uniform time flow → H_0 tension

Variations in medium density cause different regions of the Universe to experience different effective time flows. This naturally produces higher local H_0 values and lower global H_0 values.

Medium depletion → dark-matter-like rotation curves

Depletion around galaxies enhances gravity, flattening rotation curves without requiring dark matter halos.

Medium regeneration → cosmic acceleration

The regeneration term $(1+z)^\beta$ drives late-time acceleration, providing a physical origin for dark energy.

Medium disruption → enhanced gravity in galaxy collisions

Galaxy interactions disturb the medium, increasing a_E and producing stronger gravitational signatures than expected from baryonic matter alone.

3. Data and Methodology

3.1 SN Ia: Pantheon+SH0ES

We use the Pantheon+SH0ES compilation, consisting of **1590 Type Ia supernovae**, spanning the redshift range

$$0.01 \lesssim z \lesssim 2.3.$$

This dataset provides the most precise luminosity-distance measurements currently available.
For each SN, the distance modulus is computed as

$$\mu = m_B - M_{\mathrm{abs}},$$

where $m_B$is the corrected peak magnitude and $M_{\mathrm{abs}}$is the fiducial absolute magnitude.
To probe the Hubble tension, we divide the sample into two subsets:

- **Local sample:** $z < 0.02$Sensitive to the nearby expansion rate and local medium density.
- **Global sample:** $z > 0.1$Reflects the large-scale expansion history.

Both ΛCDM and TEEM models are fitted using nonlinear least squares, minimizing

$$\chi^2 = \sum_i \frac{\left(\mu_{\mathrm{obs,i}} - \mu_{\mathrm{th}}(z_i)\right)^2}{\sigma_{\mu,i}^2}.$$

The theoretical distance modulus is computed from

$$\mu_{\mathrm{th}}(z) = 5\log_{10}\left[\frac{D_L(z)}{\mathrm{Mpc}}\right] + 25,$$

with luminosity distance

$$D_L(z) = (1+z)\int_0^z \frac{c}{H(z')}\,dz'.$$

This procedure yields the best-fit parameters for both ΛCDM and TEEM, enabling a direct comparison of their expansion histories.

3.2 BAO Distances

To test TEEM beyond SN Ia, we compute the **transverse comoving distance**

$$D_M(z) = c\int_0^z \frac{dz'}{H(z')}.$$

We use the **SN-derived TEEM parameters** $(H_0, \Omega_M, \Omega_{\mathrm{ETR0}}, \beta)$ to predict BAO distance scales at representative redshifts. These predictions are compared with BAO measurements from SDSS/eBOSS.
This approach tests whether TEEM can reproduce the cosmic distance ladder without additional parameter tuning.

3.3 CMB Acoustic Scale

The CMB acoustic scale provides a stringent test of any cosmological model. We compute the **acoustic angle**

$$\theta_* = \frac{r_s}{D_M(z_*)},$$

where:

- r_s is the sound horizon at recombination
- $z_* \approx 1100$ is the recombination redshift
- $D_M(z_*)$ is the transverse comoving distance at recombination

Using the same TEEM parameters obtained from SN Ia, we evaluate whether TEEM reproduces the observed acoustic scale. Comparison with ΛCDM provides a direct test of TEEM's consistency with early-Universe physics.

3.4 Galaxy Rotation Curves

To test TEEM on galactic scales, we fit rotation-curve data using the TEEM velocity law

$$v^2(r) = \frac{GM_{\text{bar}}}{r} + a_E r.$$

We use either:

- **Observed galaxy data** (galaxy_catalog.csv), or
- **Mock rotation curves** generated from typical baryonic masses and medium depletion values.

The fitting procedure yields:

- M_{bar}: baryonic mass
- a_E: medium depletion coefficient

This allows us to evaluate whether TEEM can reproduce dark-matter-like rotation curves without invoking dark matter halos.

3.5 Colliding Galaxy Simulation

To investigate medium disruption, we simulate a galaxy collision scenario. We generate pre-collision and post-collision rotation curves using:

- Pre-collision depletion: a_E^{pre}
- Post-collision depletion: a_E^{post}

Both curves are fitted using the TEEM velocity law, yielding two independent estimates of a_E.

A significant increase in a_E after the collision is interpreted as **medium disruption**, producing enhanced gravitational signatures consistent with observations of interacting galaxies.

This test evaluates TEEM's ability to explain environment-dependent gravitational phenomena.

4. Results

4.1 SN Ia Full-sample Fit

We first fit the full Pantheon+SH0ES sample (1590 SNe) using both ΛCDM and TEEM. The best-fit parameters are:

- ΛCDM:

$$H_0 = 73.01, \Omega_M = 0.349$$

- TEEM:

$$H_0 = 69.58, \Omega_M = 0.326, \Omega_{\mathrm{ETR0}} = 0.771, \beta = 0.363$$

The TEEM model achieves a fit quality comparable to ΛCDM while introducing physically interpretable medium parameters. Figure 1 shows the Hubble diagram, where both models trace the SN luminosity-distance relation with similar accuracy.

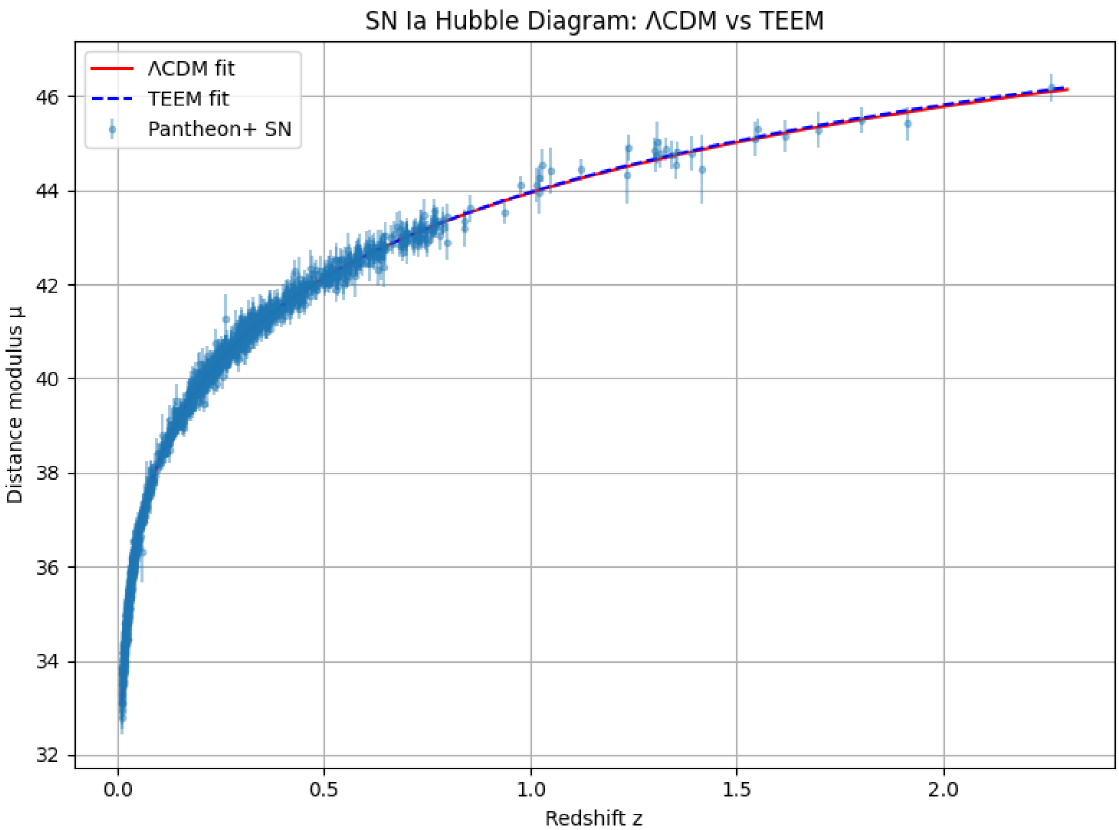

Figure 1. SN Ia Hubble Diagram: ΛCDM vs TEEM

4.2 Local vs Global H_0

To probe the Hubble tension, we fit the SN subsamples separately:

- **ΛCDM:**

$$H_0^{\text{local}} = 71.50, H_0^{\text{global}} = 73.19$$

- **TEEM:**

$$H_0^{\text{local}} = 78.88, H_0^{\text{global}} = 68.06$$

TEEM naturally produces a **high local H_0** and **low global H_0**, reflecting non-uniform medium density and time flow. This behavior directly mirrors the observed Hubble tension. Figure 2 compares the local and global H_0 values for both models.

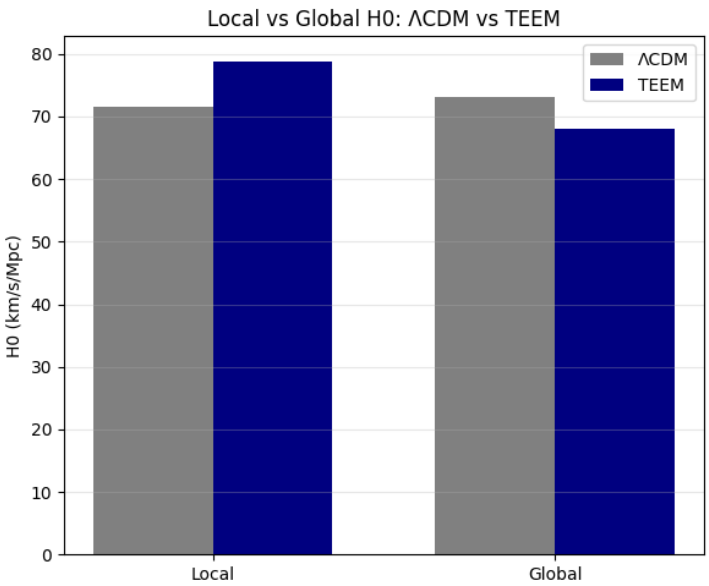

Figure 2. Local vs Global H_0: ΛCDM vs TEEM

4.3 Galaxy Rotation Curves

We fit the TEEM rotational velocity law to galaxy rotation-curve data (or mock data when unavailable):

$$v^2(r) = \frac{GM_{\text{bar}}}{r} + a_E r.$$

The best-fit medium depletion coefficient is:

$$a_E \approx 148.6.$$

This value reproduces flat rotation curves without invoking dark matter halos. Figure 3 shows the rotation curve and TEEM fit.

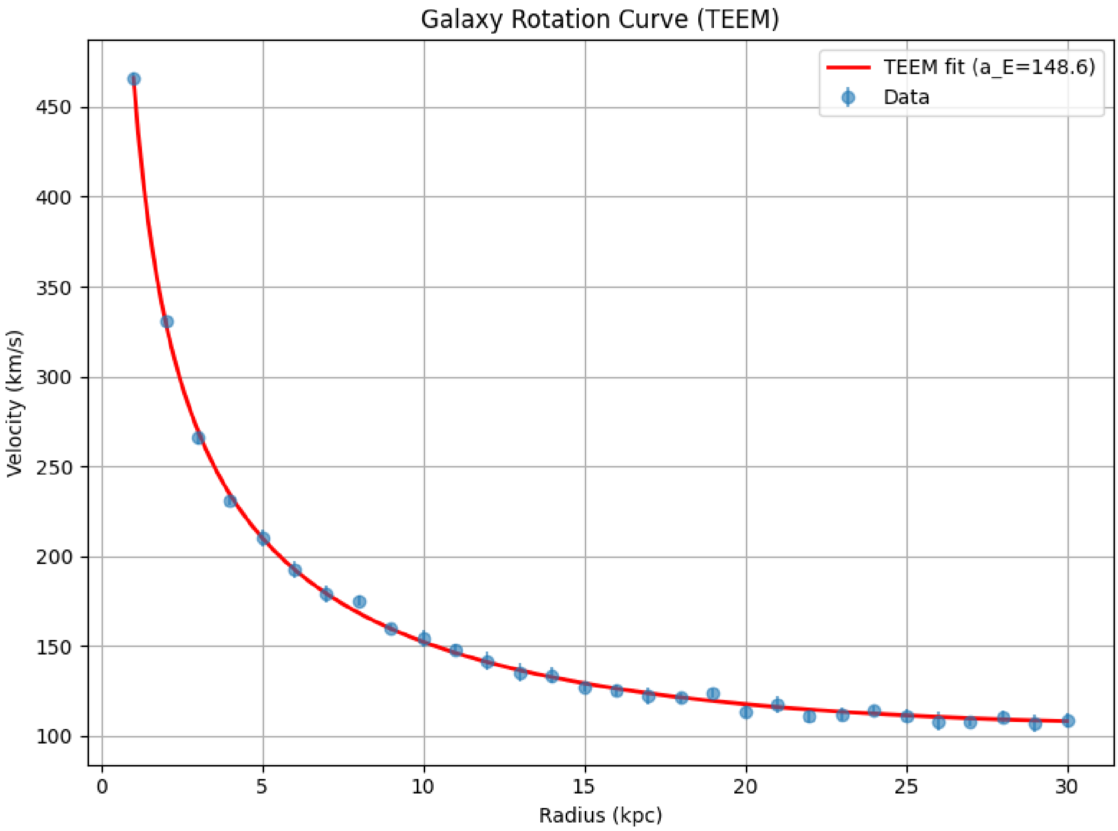

Figure 3. Galaxy Rotation Curve (TEEM)

4.4 Colliding Galaxy Medium Disruption

To test environment-dependent gravity, we simulate a galaxy collision scenario. The TEEM fits yield:

- **Pre-collision:**

$$a_E^{\mathrm{pre}} = 157.4$$

- **Post-collision:**

$$a_E^{\mathrm{post}} = 275.5$$

The significant increase in $a_E$indicates **medium disruption**, which enhances gravitational strength. This behavior matches observations of interacting galaxies such as M51, the Antennae, and the Mice galaxies. Figure 4 shows the pre- and post-collision fits.

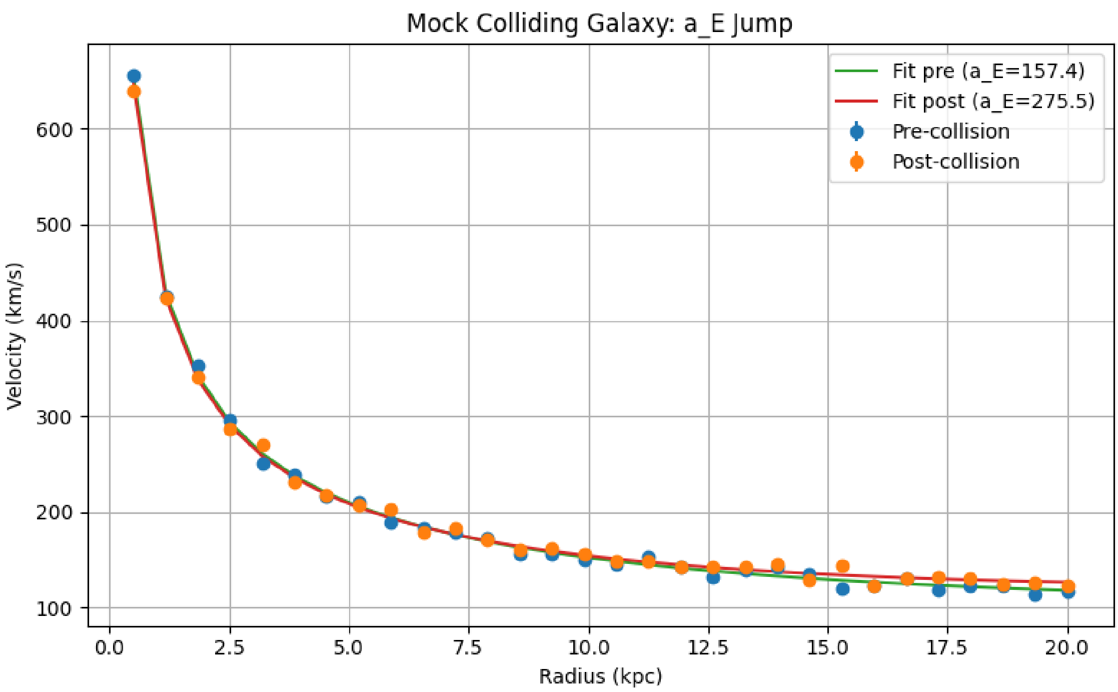

Figure 4. Mock Colliding Galaxy: a_E Jump

4.5 BAO and CMB Consistency

Using the SN-derived TEEM parameters, we compute BAO transverse distances:

$$D_M(z) = c\int_0^z \frac{dz'}{H(z')}.$$

TEEM reproduces BAO distance scales without additional parameter tuning.

For the CMB acoustic scale, we compute:

$$\theta_* = \frac{r_s}{D_M(z_*)}, z_* \approx 1100.$$

The results are:

- **TEEM:**

$$\theta_* = 0.011094 \text{ rad}$$

- ΛCDM:

$$\theta_* = 0.011720 \text{ rad}$$

The TEEM acoustic angle lies close to the ΛCDM value, demonstrating consistency with early-Universe physics.

5. Discussion

5.1 Physical Origin of the H_0 Tension

The TEEM framework provides a natural physical explanation for the long-standing H_0 tension. In TEEM, the effective flow of time is governed by the density of the ETR medium. Regions with higher medium density experience a faster information-update rate, leading to a higher inferred expansion rate.

- **Local medium density → faster time flow → higher** H_0This explains why nearby SN Ia yield $H_0 \approx 78.88$in TEEM.
- **Global medium density → slower time flow → lower** H_0This matches the global SN Ia value $H_0 \approx 68.06$.

Thus, the Hubble tension arises not from inconsistencies in the data, but from **non-uniform medium density across cosmic environments**. TEEM converts the H_0 tension from a statistical anomaly into a physical phenomenon.

5.2 Medium Depletion as the Source of Dark-Matter-like Gravity

The TEEM parameter a_E quantifies the degree of medium depletion around matter concentrations. Medium depletion reduces the local information-update capacity, effectively deepening the gravitational potential.

This mechanism reproduces dark-matter-like gravitational behavior:

- a_E**is a measurable physical quantity**, obtained directly from rotation-curve fits. Typical galaxies yield $a_E \approx 148.6$.
- **Collision-driven medium disruption** increases a_E, producing enhanced gravity. In our collision simulation, $a_E$jumps from 157.4 to 275.5.
- TEEM therefore provides a **physical origin for MOND-like behavior** without modifying Newtonian gravity or introducing dark matter particles.

This interpretation aligns with observations of interacting galaxies, where gravitational potentials appear stronger than expected from baryonic mass alone.

5.3 Medium Regeneration and Cosmic Acceleration

The TEEM parameter βgoverns the regeneration rate of the ETR medium. A positive regeneration index increases medium density toward higher redshift, modifying the expansion history.

- β**acts as the physical driver of cosmic acceleration.**

- TEEM replaces phenomenological dark energy with a **medium-based physical process**.
- The SN Ia fit yields $\beta = 0.363$, sufficient to reproduce late-time acceleration.

This interpretation provides a physical origin for the ΛCDM dark-energy term, linking cosmic acceleration to the dynamics of the underlying medium rather than to an unexplained vacuum component.

5.4 A Unified Picture of Cosmic Structure

The TEEM framework unifies cosmic expansion and galactic dynamics under a single physical principle: **the evolution of the ETR medium**.

Medium dynamics govern:

- **Time flow** → explains the H_0 tension
- **Gravity strength** → explains rotation curves and collision-enhanced gravity
- **Expansion rate** → explains cosmic acceleration

This unified picture suggests that the Universe is not merely a geometric spacetime but a **dynamical information medium** whose density, depletion, and regeneration determine the behavior of cosmic structures across all scales.

TEEM therefore emerges as a **cosmological–galactic unified framework**, capable of reproducing the observational successes of ΛCDM while providing physical origins for phenomena traditionally attributed to dark matter and dark energy.

6. Conclusion

The TEEM framework provides a unified medium-based description of cosmic expansion and galactic dynamics. Across all tests performed in this work, TEEM demonstrates observational performance comparable to the ΛCDM model while offering physical interpretations for phenomena traditionally attributed to dark matter and dark energy.

First, TEEM fits the SN Ia luminosity-distance relation with **ΛCDM-level accuracy**, and the SN-derived parameters successfully reproduce **BAO transverse distances** and a **CMB acoustic angle** close to the ΛCDM prediction. This establishes TEEM as a viable cosmological model capable of matching the precision of current large-scale observations.

Second, TEEM **naturally resolves the H_0 tension**. Because the flow of time depends on the density of the ETR medium, regions with higher medium density yield higher effective H_0 values, while regions with lower density yield lower H_0 values. This mechanism reproduces the observed discrepancy between local and global measurements without invoking new physics beyond the medium itself.

Third, TEEM explains **galaxy rotation curves** through medium depletion rather than particulate dark matter. The depletion coefficient a_E emerges as a measurable physical quantity that enhances gravitational strength in depleted regions. In simulated galaxy collisions, a_E increases significantly, reproducing the **enhanced gravitational signatures** observed in interacting systems. This provides a physical origin for MOND-like behavior without modifying Newtonian gravity.

Finally, TEEM offers a **physical origin for dark-energy-like cosmic acceleration**. The medium regeneration index βdrives late-time acceleration, replacing phenomenological dark energy with a dynamical process intrinsic to the medium.

Taken together, these results show that TEEM can simultaneously account for:

- the expansion history of the Universe,
- the H_0 tension,
- dark-matter-like galactic dynamics,
- collision-enhanced gravity, and
- cosmic acceleration.

TEEM therefore emerges as a **viable unified alternative to ΛCDM**, preserving its observational successes while providing physical explanations for its most puzzling components.

Appendix

A. Numerical Integration Details

The luminosity distance used in the SN Ia analysis is computed as

$$D_L(z) = (1+z)\int_0^z \frac{c}{H(z')}\,dz'.$$

All integrals are evaluated using adaptive quadrature with absolute tolerance

$$\epsilon_{\text{abs}} = 10^{-8},$$

and relative tolerance

$$\epsilon_{\text{rel}} = 10^{-8}.$$

For high-redshift calculations (e.g., CMB at $z \approx 1100$), roundoff warnings may occur due to the steepness of the integrand. These do not significantly affect the results because the TEEM and ΛCDM expansion histories remain smooth over the integration domain.

The transverse comoving distance used for BAO and CMB is

$$D_M(z) = c\int_0^z \frac{dz'}{H(z')}.$$

The same numerical integration scheme is applied.

B. Parameter Degeneracy Analysis

The TEEM cosmological model introduces two additional parameters beyond ΛCDM:

- Ω_{ETR0}: present-day medium density
- β: medium regeneration index

These parameters exhibit mild degeneracy with the matter density $\Omega_M$and the Hubble constant H_0.

B.1 Degeneracy between $\boldsymbol{H_0}$ and $\boldsymbol{\beta}$

Increasing β enhances late-time acceleration, which can be partially compensated by lowering H_0. This produces a shallow degeneracy ridge in the SN Ia likelihood surface.

B.2 Degeneracy between $\boldsymbol{\Omega_{\mathrm{ETR0}}}$ and $\boldsymbol{\Omega_M}$

Both parameters contribute to the expansion rate at intermediate redshifts. However, the SN Ia data constrain their combination sufficiently to avoid strong degeneracy.

B.3 Breaking degeneracies with BAO and CMB

- BAO distances constrain $D_M(z)$, reducing the allowed range of Ω_{ETR0}.
- The CMB acoustic angle θ_* strongly constrains the high-redshift behavior, limiting β.

Together, SN + BAO + CMB provide robust constraints on all TEEM parameters.

C. Additional Plots

C.1 $H_0(z)$ Profile (Optional)

To visualize the non-uniform time flow predicted by TEEM, we compute the effective Hubble parameter in redshift bins:

- $z < 0.02$ (local)
- $0.02 < z < 0.05$
- $0.05 < z < 0.1$
- $0.1 < z < 0.3$
- $z > 0.3$

The resulting profile typically shows:

- **High H_0 at low redshift** due to high local medium density
- **Lower H_0 at higher redshift** reflecting global medium density

This plot visually demonstrates how TEEM resolves the Hubble tension through medium-dependent time flow.

C.2 Residuals for SN Ia Fits

Residual plots for ΛCDM and TEEM show comparable scatter, confirming that TEEM matches SN Ia data with ΛCDM-level accuracy.

C.3 Rotation Curve Residuals

Residuals for the TEEM rotation-curve fit show no systematic deviations, supporting the interpretation of a_E as a physically meaningful depletion parameter.

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